Brent M. Engelman

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PROFESSIONAL SUMMARY

Solutions Architect and Lead QA Engineer with over a decade of experience designing test automation, CI/CD pipelines, API integrations, and database-driven systems across eCommerce, government/legal, and defense environments. Built GitHub Actions pipelines with Mabl and Selenium WebDriver, and recovered $700,000 to $1,000,000 in unbilled licensing revenue through analytics and SQL configuration. Skilled in translating requirements into testable technical solutions, troubleshooting complex systems, and documenting operational standards.

TECHNICAL SKILLS

- **Automation & AI Integration**: Custom CI/CD Workflows, GitHub Actions (Automated Triggers/Re-runs), Low-Code/No-Code Platform Integration, Process Automation, AI-Assisted Development Tools, Webhooks & Event-Driven Triggers
- **Software & API Development**: Python, JavaScript, C#, C++, PHP, HTML/CSS, PowerShell, RESTful APIs, JSON, Microservices & Third-Party System Integration, Object-Oriented Programming (OOP), Java
- **Databases & Data Engineering**: Relational Database Architecture (SQL Server, PostgreSQL), SQL Query Optimization, Data Modeling, ETL/Data Flows, Heap Analytics Data Tracking
- **Systems Architecture & Security**: Cloud Modernization, AWS Infrastructure, IIS & EWS Configuration, PII Privacy Governance, CJIS Compliance, Security Controls & Penetration Concepts (Metasploit, Wireshark, SIPR/NIPR)
- **Testing & Build Optimization**: Playwright, Selenium WebDriver, Mabl Automation Framework, Hybrid Build/Update Tools- , Standardized Testing Pipelines, Test Automation, Test Automation Framework Design, Software Quality Assurance, TestNG, JUnit
- **Vendor & Stakeholder Leadership**: Technical Vendor Management, Strategic Sourcing, SME Training & Capability Building, User Story & Functional Requirements Engineering
- **Software Development Methodologies**: Agile and Scrum

PROFESSIONAL EXPERIENCE

Nomad eCommerce

Solutions Architect / Lead QA Engineer — *O'Fallon, IL* • **May 2024 - Sep 2026**

- System & API Integration: Engineered end-to-end integration workflows, webhooks, and RESTful service connections between multi-tenant eCommerce applications and third-party systems, managing vendor relationships to enable scalable partner capabilities.
- Workflow & Build Automation: Built custom build and update tools to transition manual deployment procedures into rapid hybrid automation pipelines, significantly accelerating update deployment cycles and feature delivery.
- Requirements to Architecture: Translated unstructured client feature requests and high-level concepts into technical specifications, actionable development stories, API interface specifications, and database configuration models.
- Vendor Management & Sourcing: Advised executive leadership on resource allocation and third-party vendor management, strategically outsourcing non-core development tasks to external partners while maintaining internal focused engineering capacity.
- System Security & Documentation: Authored technical SOPs and implementation guides for complex system security modules, including CAPTCHA protection, payment gateway processor integrations, and authentication protocols.

Karpel Solutions

QA Engineer — *St. Louis, MO* • **Mar 2019 - May 2024**

- Database Analytics & Revenue Recovery: Configured data tracking architectures by integrating Heap Analytics into core codebase environments via custom scripts and SQL configurations, identifying tracking variances that recovered $700,000 to $1,000,000 in unbilled licensing revenue.
- CI/CD Pipeline & Test Automation: Designed test automation frameworks and deployed automated CI/CD test pipelines using GitHub Actions, including scheduled triggers, branch deployment hooks, and automated failure re-runs, alongside Mabl and Selenium WebDriver to stabilize application builds.
- Data Modeling & Cloud Transition: Spearheaded technical cloud modernization projects, migrating legacy government and legal sector clients from physical record systems to secure, cloud-hosted relational database architectures.
- Data Privacy & Compliance Controls: Formulated and enforced PII data privacy internal controls within mission-critical case management software to maintain strict legal and CJIS regulatory compliance without compromising performance.
- Cross-Platform UI/UX Optimization: Performed software quality assurance by identifying and remediating OS- and browser-level code variances across macOS and Windows environments, enabling front-end developers to engineer responsive, cross-platform user experiences.

IT Support Analyst (System Administrator) — *St. Louis, MO* • **Jun 2017 - Mar 2019**

- Infrastructure Troubleshooting: Diagnosed and resolved complex server-side and database issues involving Exchange Web Services (EWS), SQL Server queries, IIS web servers, and local network configurations.
- Workforce Capability Building: Designed and delivered technical training programs for organizational Subject Matter Experts (SMEs), establishing baseline procedures for system administration, software update rollouts, and security best practices.

PhasedLogix

Helpdesk Analyst *St. Louis, MO* • **Oct 2016 - Feb 2017**

- Provided tier-1/tier-2 technical support, network troubleshooting (IPv4, DNS, TCP/UDP), and system configuration for external business clients.

United States Navy

Cryptologic Technician (Active Duty)

- Mission-Critical Technical Operations: Managed high-security communications channels, satellite tracking software, server hardware, and fiber-optic network infrastructure in high-pressure operational settings.
- System Status Briefings: Prepared concise technical reports, risk assessments, and system status briefings for commanding officers to maintain continuous fleet operational readiness.

EDUCATION

Vatterott College

Information Systems Security Specialist (ISSP) *Fairview Heights, IL*

Southwestern Illinois College

Associate of Science, Mathematics *Belleville, IL*

PROFESSIONAL CERTIFICATIONS

- AWS Certified Cloud Practitioner: Amazon Web Services
- MTA Database Fundamentals: Microsoft Certified
- MTA Database Administration Fundamentals: Microsoft Certified
- Project+: CompTIA Certifications
- ITIL Foundation Certificate in IT Service Management